Digital Payment Adoption & Fraud Risk: A Quasi A/B Test Analysis

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1. Project Overview

A FinTech company introduced a digital-first payment strategy to improve transaction efficiency and reduce fraud exposure. This study uses a quasi-experimental A/B testing approach to evaluate whether users on modern payment channels (Mobile & ECOM) demonstrate significantly better performance outcomes compared to legacy channels (Web, POS, Internet Banking), based on fraud rate, transaction value, and risk behavior metrics.

2. Methodology

This project was executed using a rigorous analytical workflow in Python:

- **Data Cleaning:** Prepared raw transaction data for high-fidelity analysis.
- **Exploratory Data Analysis (EDA):** Uncovered core trends and patterns in channel performance.
- **Hypothesis Testing:** Conducted statistical validation to ensure findings were reliable.

3. Statistical Tests Conducted

- **Independent Two-Sample t-Test:** Compared average transaction values between groups, confirming significant differences in spending behavior.
- **Two-Proportion Z-Test:** Analyzed fraud rate differences. While Digital (0.315%) was slightly higher than Legacy (0.285%), the impact remains small in business terms.
- **Cohen's d (Effect Size):** Validated that the practical impact of observed differences is relatively small.
- **Confidence Interval Analysis:** Provided certainty around the fraud rate findings.

4. Dashboard Summary

Key Metrics

Metric	Value
Total Transactions	1,000,000
Total Fraud Transactions	3,000

Overall Fraud Rate	0.30%
Avg. Transaction Amount	$156,951.42
Avg. Merchant Risk Score	0.3956

A/B Test Results

- **Fraud Rate:** Legacy (0.285%) vs. Digital (0.315%). Note: A slight increase with minimal business impact.
- **Avg. Value:** Legacy (195,801) vs. Digital (118,063).
- **Risk Score:** Legacy (0.1581) vs. Digital (0.1445). Digital has lower risk.

5. Recommendations

- Continue digital payment expansion.
- Deploy adaptive, machine learning-based fraud detection.
- Focus on risk-based authentication to streamline user experience.
- Maintain real-time monitoring for high-risk behaviors.
- Strengthen fraud controls specifically for high-risk transactions.